Q1 Pick out the expression of electric potential energy from the following.

(A) $U = \frac{1}{(4\pi\varepsilon_o)} \times \left[\frac{q_1 q_2}{r}\right]$

(B) $U = 1 \times (4\pi\varepsilon_0) \times \left[\frac{q_1 q_2}{r}\right]$

(C) $U = \frac{1}{(4\pi\varepsilon_o)} \times \left[\frac{q_1}{q_2} r\right]$

(D) $U = \frac{1}{(4\pi\varepsilon_o)} \times [q_1 q_2]$

Q2 Two isolated metallic spheres, one with a radius R and another with a radius 5R, each carries a charge 'q' uniformly distributed over the entire surface. Which sphere stores more electric potential energy?

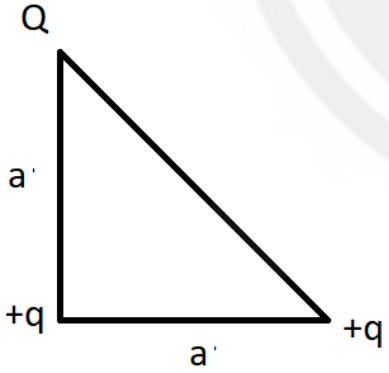
(A) The sphere with radius 5R

(B) Both of the spheres will have the same energy

(C) The sphere with radius R

(D) Initially it will be the sphere with radius 5R then it will shift to the sphere with radius R

Q3 What should be the value of Q so that the potential energy of the system is zero?



(A) 2q

(B) $\frac{-q}{2}$

(C) $\sqrt{2}q$

(D) $\frac{-2q}{(2+\sqrt{2})}$

Q4 Three point charges q, $-2q$, and q are placed at the corners of an equilateral triangle of side a. The electrostatic potential energy of the system is:

(A) $\frac{kq^2}{a}$

(B) $-\frac{3kq^2}{a}$

(C) $-\frac{kq^2}{a}$

(D) Zero

Q5 A positively charged particle is released from rest in a uniform electric field. The electric potential energy of the charge:

(A) Remains constant because the electric field is uniform

(B) Increases because the charge moves along the electric field

(C) Decreases because the charge moves along the electric field

(D) Decreases because the charge moves opposite to the electric field

Q6 When one electron is taken towards the other electron, then the electric potential energy of the

(A) decreases

(B) increases

(C) remains unchanged

(D) will become zero

Q7 An electric dipole of moment $\vec{p}$ is placed normal to the lines of force of electric intensity $\vec{E}$, then the work done in deflecting it through an angle of $180°$ is

(A) pE

(B) $+2pE$

(C) $-2pE$

(D) zero

Q8 An electric dipole of length $1\ \text{cm}$ is placed with the axis making an angle of $30°$ to an electric field of strength $10^4\ \text{NC}^{-1}$. If it experiences a torque of $10\sqrt{2}\ \text{Nm}$, the potential energy of the dipole is

(A) $0.245\ \text{J}$

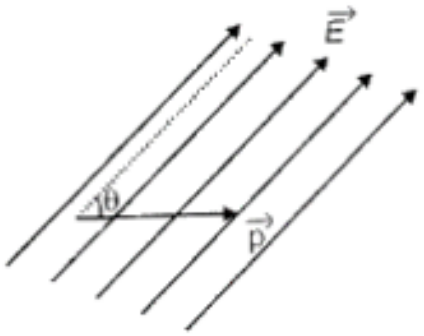
(B) $2.45\ \text{J}$

(C) $0.0245\ \text{J}$

(D) $24.5\ \text{J}$

Q9 A dipole is placed in constant electric field E as shown. If it is released from rest from this

position, find the kinetic energy gained when it is parallel to the electric field vector. (Assume dipole is small and placed in a gravity free space)



(A) $-PE(1-\cos\theta)$
(B) $PE(1-\cos\theta)$
(C) $PE(1-\sin\theta)$
(D) $-PE(1-\sin\theta)$

Q10 Polar molecules are the molecules:
(A) Acquire a dipole moment only in the presence of electric field due to displacement of charges.
(B) Acquire a dipole moment only when magnetic field is absent.
(C) Having a permanent electric dipole moment.
(D) Having zero dipole moment.

Q11 The molecules having permanent electric dipole moment even in the absence of electric field are called as:
(A) Non polar molecules.
(B) Polar molecules.
(C) Both (a) and (b).
(D) None.

Q12 Statement-1 : If temperature is increased, the dielectric constant of a polar dielectric decreases whereas that of a non-polar dielectric does not change significantly.
Statement-2 : The magnitude of dipole moment of individual polar molecule decreases significantly with increase in temperature.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement- 2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false statement-2 is true

Q13 The total dipole moment of dielectric with polar molecules in absence of an external electric field is
(A) zero (B) negative
(C) infinite (D) None of these

Q14 Dielectric constant for a metal is:
(A) zero (B) infinite
(C) 1 (D) 10

Answer Key

Q1	A
Q2	C
Q3	D
Q4	B
Q5	C
Q6	B
Q7	D
Q8	D
Q9	B
Q10	C
Q11	B
Q12	C
Q13	A
Q14	B

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Explanation: electric potential energy of a system of charges is the total amount of work done in bringing the various charges to their respective positions from infinitely large mutual separations. The expression for electric potential energy is given by:

$$U = \frac{1}{(4\pi\varepsilon_o)} \times \left[\frac{q_1 q_2}{r}\right]$$

Video Solution:

Q2 Text Solution:

Explanation: The sphere with radius R stores more electric potential energy. According to the electric potential energy equation

$\rightarrow U = \frac{1}{(4\pi\varepsilon_o)} \times \left[\frac{q_1 q_2}{r}\right]$ Potential energy is inversely proportional to radius. Therefore, the sphere with lesser radius will store more energy. So, the smaller sphere will store more energy.

Video Solution:

Q3 Text Solution:

Explanation: The distance between the charges Q and + q is $\sqrt{2}a$ a according to Pythagoras theorem. Therefore, the total potential energy of the system becomes $\left(Q \times \frac{q}{a}\right)+\left(q \times \frac{q}{a}\right)+\left(Q \times \frac{q}{\sqrt{2}a}\right)$. But according to the questions, the total potential energy is zero, i.e. $\left(Q \times \frac{q}{a}\right)+\left(q \times \frac{q}{a}\right)+\left(Q \times \frac{q}{\sqrt{2}a}\right)=0$.

Therefore, simplifying the equation, we get Q $= \frac{-2q}{(2+\sqrt{2})}$.

Video Solution:

Q4 Text Solution:

Total potential energy = sum of interaction energies:

$$U = k\left(\frac{q\times(-2q)}{a} + \frac{q\times q}{a} + \frac{q\times(-2q)}{a}\right) = -\frac{3kq^2}{a}$$

Video Solution:

Q5 Text Solution:

The electric potential energy of the positively charged particle decreases as it moves in the direction of the electric field.

Video Solution:

Q6 Text Solution:

Potential energy of system will be given by $\frac{(-e)(-e)}{4\pi\epsilon_0 r} = \frac{e^2}{4\pi\epsilon_0 r} r$ decrease then potential energy will be increases.

Video Solution:



Q7 Text Solution:

The potential energy of a dipole in a uniform field is

$$U = -pE\cos\theta,$$

where θ is the angle between $\mathbf{p}$ and $\mathbf{E}$.

- Initially the dipole is **normal** to the field: $\theta_i = 90^\circ \Rightarrow U_i = -pE\cos 90^\circ = 0.$
- After it is turned through 180° the dipole is still normal to the field (now pointing the other way): $\theta_f = 90^\circ + 180^\circ = 270^\circ \Rightarrow U_f =. -pE\cos 270^\circ = 0$

Work done by an external agent equals the change in potential energy:

$$W = U_f - U_i = 0 - 0 = 0.$$

Hence no net work is required.

Video Solution:



Q8 Text Solution:

$$\tau = pE\sin\theta \quad \Rightarrow$$

$$p = \frac{\tau}{E\sin\theta} = \frac{10\sqrt{2}}{(10^4)(\frac{1}{2})}$$

$$= \frac{2\sqrt{2}}{10^3}\ \mathrm{C\backslash cdotpm} \approx 2.8 \times 10^{-3}\ \mathrm{C\backslash cdotpm.}$$

Potential energy of a dipole in a uniform field: $U = -pE\cos\theta$

$$U = -\left(2.8 \times 10^{-3}\right)\left(10^4\right)\cos 30^\circ \approx -24.5\ \mathrm{J}.$$

Magnitude $= 24.5\ \mathrm{J}$

Video Solution:



Q9 Text Solution:

Key concept: Work-energy theorem

$U = -\vec{P}\cdot\vec{E}$

$W = \Delta k = -[\Delta U] = -[U_f - U_i]$

$k_f - k_i = -[U_f - U_i] =$

$-[-PE + PE\cos\theta]$

$=[PE - PE\cos\theta]$

$= PE(1 - \cos\theta)$

Video Solution:



Q10 Text Solution:

Polar molecules have centres of positive and negative charges apart by a certain distance which means positive and negative polarity

arises. So, the polar molecules have permanent dipole moment.

Video Solution:

Q11 **Text Solution:**

Molecules having permanent electric dipole moment even in the absence of electric field are called as polar molecules.

Video Solution:

Q12 **Text Solution:**

Answer : C

- **Statement-1 – True.** Temperature agitates permanent dipoles in a *polar* dielectric, reducing their net alignment, so K (dielectric constant) falls noticeably; for *non-polar* dielectrics K is dominated by electronic/atomic polarisation and is almost temperature-independent.
- **Statement-2 – False.** The **intrinsic** dipole moment of a molecule is set by its bond geometry and does **not** shrink with temperature; only the orientation (hence bulk polarisation) changes.

Video Solution:

Q13 **Text Solution:**

Due to random orientation of dipole moments, the total dipole moment is zero in the absence of external field.

Video Solution:

Q14 **Text Solution:**

Permittivity of metals is very high relative to the permittivity of free space.
Thus, dielectric constant for metal is infinite

Video Solution: